

Quadratic Forms, Nested Models and Hypothesis Testing

A Crash Course from First Principles

Linear Statistical Models — Lectures 7–10 (10 Aug – 24 Aug)

The third of four crash courses. Crash Course 2 told you *what* to estimate and that $a^T \hat{\beta}$ is best. This one tells you how to *test* whether a smaller model would do, which needs the distribution theory of quadratic forms $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$. The F -statistic of ANOVA is the destination.

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1 Why this exists

Every test in this course is the same question in different clothes:

The one question

You have a big model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Someone proposes a smaller model — drop some predictors, or impose $L^T\boldsymbol{\beta} = \boldsymbol{\theta}_0$. **Is the small model good enough?**

Fitting the small model must give a larger residual sum of squares (fewer directions to project onto). The test asks whether that increase is *more than noise would explain*:

$$F = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/(\text{d.f. lost})}{\text{SSE}_{\text{full}}/(n - \rho(\mathbf{X}))}.$$

Everything below is the machinery that makes this exact.

2 Nested models and the projection lemma

2.1 Setup

Full model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, sub-model $\mathbf{Y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$, where $\mathbf{Z}$ is a subset of the columns of $\mathbf{X}$:

$$\mathbf{X}_{n \times p} = [\mathbf{Z}_{n \times q} : \mathbf{W}_{n \times s}], \quad \boldsymbol{\beta} = \begin{pmatrix} \boldsymbol{\gamma} \\ \boldsymbol{\delta} \end{pmatrix}, \quad p = q + s.$$

Then $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X})$: the sub-model is **nested** in the full model, and $\mathbf{Y} = \mathbf{Z}\boldsymbol{\gamma} + \mathbf{W}\boldsymbol{\delta} + \boldsymbol{\varepsilon}$ with the sub-model being $\boldsymbol{\delta} = \mathbf{0}$.

2.2 The lemma that runs everything

Projection decomposition lemma

$$P_{\mathbf{X}} = P_{\mathbf{Z}} + P_{\mathbf{W}|\mathbf{Z}}$$

where $P_{\mathbf{W}|\mathbf{Z}}$ is the orthogonal projector onto $\mathcal{C}((I - P_{\mathbf{Z}})\mathbf{W})$ — the part of $\mathbf{W}$ left after removing everything explainable by $\mathbf{Z}$.

Proof. Let $u \in \mathcal{C}(\mathbf{X})$, so $u = \mathbf{Z}v_1 + \mathbf{W}v_2$. Then $(I - P_{\mathbf{Z}})u = (I - P_{\mathbf{Z}})\mathbf{W}v_2$, since $(I - P_{\mathbf{Z}})\mathbf{Z}v_1 = \mathbf{0}$. Also $u = P_{\mathbf{Z}}u + (I - P_{\mathbf{Z}})u$. It remains to check $(I - P_{\mathbf{Z}})\mathbf{W}v_2 = P_{\mathbf{W}|\mathbf{Z}}u$. Writing $M = (I - P_{\mathbf{Z}})\mathbf{W}$ and using $(I - P_{\mathbf{Z}})^2 = I - P_{\mathbf{Z}}$:

$$P_{\mathbf{W}|\mathbf{Z}}u = M(M^T M)^{-1}M^T u = (I - P_{\mathbf{Z}})\mathbf{W}(\mathbf{W}^T(I - P_{\mathbf{Z}})\mathbf{W})^{-1}\mathbf{W}^T(I - P_{\mathbf{Z}})\mathbf{W}v_2 = (I - P_{\mathbf{Z}})\mathbf{W}v_2.$$

So $u = P_{\mathbf{Z}}u + P_{\mathbf{W}|\mathbf{Z}}u$ for every $u \in \mathcal{C}(\mathbf{X})$, and both sides kill $\mathcal{C}(\mathbf{X})^\perp$. ■

What the lemma is saying

$\mathcal{C}(\mathbf{X})$ splits into two *orthogonal* pieces: the part $\mathbf{Z}$ already explains, and the extra part $\mathbf{W}$ contributes on top of $\mathbf{Z}$. Every sum-of-squares decomposition in this course is this one sentence.

2.3 The orthogonal special case, and the general formula

If $\mathbf{Z}^T\mathbf{W} = \mathbf{0}$, the columns of $\mathbf{Z}$ and $\mathbf{W}$ are already orthogonal and the criterion splits cleanly:

$$\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \|(I - P_{\mathbf{X}})\mathbf{Y}\|^2 + \|P_{\mathbf{Z}}\mathbf{Y} - \mathbf{Z}\boldsymbol{\gamma}\|^2 + \|P_{\mathbf{W}}\mathbf{Y} - \mathbf{W}\boldsymbol{\delta}\|^2.$$

Minimising each term separately shows $\hat{\boldsymbol{\gamma}}$ from the full model *equals* the estimate from the sub-model alone. Fitting the pieces separately is legitimate.

In general $\mathbf{Z}^T\mathbf{W} \neq \mathbf{0}$, and the full-model estimates are

The partial-out formulas

$$\hat{\gamma} = (\mathbf{Z}^T(I - P_{\mathbf{W}})\mathbf{Z})^{-1}\mathbf{Z}^T(I - P_{\mathbf{W}})\mathbf{Y}, \quad \hat{\delta} = (\mathbf{W}^T(I - P_{\mathbf{Z}})\mathbf{W})^{-1}\mathbf{W}^T(I - P_{\mathbf{Z}})\mathbf{Y}.$$

Compare the *sub-model* estimates $\tilde{\gamma} = (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{Y}$ and $\tilde{\delta} = (\mathbf{W}^T\mathbf{W})^{-1}\mathbf{W}^T\mathbf{Y}$: in general $\hat{\gamma} \neq \tilde{\gamma}$.

Derivation of $\hat{\delta}$. From $\mathbf{X}\hat{\beta} = P_{\mathbf{X}}\mathbf{Y} = P_{\mathbf{Z}}\mathbf{Y} + P_{\mathbf{W}|\mathbf{Z}}\mathbf{Y}$ and $\mathbf{X}\hat{\beta} = \mathbf{Z}\hat{\gamma} + \mathbf{W}\hat{\delta}$, multiply by $(I - P_{\mathbf{Z}})$:

$$(I - P_{\mathbf{Z}})\mathbf{W}\hat{\delta} = (I - P_{\mathbf{Z}})P_{\mathbf{W}|\mathbf{Z}}\mathbf{Y} = P_{\mathbf{W}|\mathbf{Z}}\mathbf{Y}.$$

Multiply by $\mathbf{W}^T$ and use $(I - P_{\mathbf{Z}})^2 = I - P_{\mathbf{Z}}$:

$$\mathbf{W}^T(I - P_{\mathbf{Z}})\mathbf{W}\hat{\delta} = \mathbf{W}^T(I - P_{\mathbf{Z}})\mathbf{Y} \implies \hat{\delta} = (\mathbf{W}^T(I - P_{\mathbf{Z}})\mathbf{W})^{-1}\mathbf{W}^T(I - P_{\mathbf{Z}})\mathbf{Y}. \quad \square$$

The Frisch–Waugh reading: “regress out, then regress”

$\hat{\delta}$ is exactly the LS estimate from the *transformed* model

$$(I - P_{\mathbf{Z}})\mathbf{Y} = (I - P_{\mathbf{Z}})\mathbf{W}\delta + \varepsilon''.$$

That is: strip $\mathbf{Z}$ ’s influence out of *both* the response and the predictors $\mathbf{W}$, then run an ordinary regression of the leftovers. This is the meaning of “the effect of $\mathbf{W}$ *adjusted for* $\mathbf{Z}$ ”.

2.4 Two consequences you will be asked for

Frisch–Waugh is not just a formula for $\hat{\delta}$; two of its specialisations are standard exam questions.

(a) The variance of a single coefficient. Take $\mathbf{Z} = \mathbf{X}_{-j}$ (all columns except the j -th) and $\mathbf{W} = \mathbf{X}_{\cdot j}$ (that one column, with scalar coefficient β_j). Write

$$d_j := \mathbf{X}_{\cdot j}^T(I - P_{-j})\mathbf{X}_{\cdot j} = \|(I - P_{-j})\mathbf{X}_{\cdot j}\|^2,$$

where P_{-j} is the orthogonal projector onto $\mathcal{C}(\mathbf{X}_{-j})$.

Variance of $\hat{\beta}_j$ (2024 exam, 4 marks)

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\mathbf{X}_{\cdot j}^T(I - P_{-j})\mathbf{X}_{\cdot j}} = \frac{\sigma^2}{d_j}$$

Proof. Frisch–Waugh gives $\hat{\beta}_j = d_j^{-1}\mathbf{X}_{\cdot j}^T(I - P_{-j})\mathbf{Y} = a^T\mathbf{Y}$ with $a = (I - P_{-j})\mathbf{X}_{\cdot j}/d_j$. Since $\text{Var}(\mathbf{Y}) = \sigma^2 I$ and $I - P_{-j}$ is symmetric idempotent,

$$\text{Var}(\hat{\beta}_j) = \sigma^2 a^T a = \frac{\sigma^2}{d_j^2} \mathbf{X}_{\cdot j}^T(I - P_{-j})(I - P_{-j})\mathbf{X}_{\cdot j} = \frac{\sigma^2}{d_j^2} \cdot d_j = \frac{\sigma^2}{d_j}. \quad \blacksquare$$

Second route: the Schur complement

$\text{Var}(\hat{\beta}_j) = \sigma^2[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}$. Permute column j to the end and use the partitioned inverse: the bottom-right entry of the inverse is the reciprocal of the Schur complement $\mathbf{X}_{\cdot j}^T\mathbf{X}_{\cdot j} -$

$\mathbf{X}_{-j}^T \mathbf{X}_{-j} (\mathbf{X}_{-j}^T \mathbf{X}_{-j})^{-1} \mathbf{X}_{-j}^T \mathbf{X}_{-j} = \mathbf{X}_{-j}^T (I - P_{-j}) \mathbf{X}_{-j} = d_j$. Either route is acceptable; the Frisch–Waugh one is shorter.

(b) **The t -statistic, and why collinearity kills power.** From (a), $\hat{\beta}_j \sim \mathcal{N}(\beta_j, \sigma^2/d_j)$. Standardising, $W_j := (\hat{\beta}_j - \beta_j) \sqrt{d_j}/\sigma \sim \mathcal{N}(0, 1)$, and inserting σ/σ into $T_j = \hat{\beta}_j/(\hat{\sigma}/\sqrt{d_j})$:

Decomposition of the t -statistic

$$T_j = \frac{\hat{\beta}_j}{\hat{\sigma}/\sqrt{d_j}} = \frac{1}{\hat{\sigma}/\sigma} \left(\frac{\beta_j}{\sigma} \sqrt{d_j} + W_j \right), \quad W_j \sim \mathcal{N}(0, 1) \text{ independent of } \hat{\sigma}^2.$$

Under $H_0 : \beta_j = 0$ this is $W_j/\sqrt{\hat{\sigma}^2/\sigma^2} \sim t_{n-p}$.

Common confusion: collinearity does not bias anything — it destroys power

Suppose the truth is $\mathbb{E}\mathbf{Y} = \beta_{j_0} \mathbf{X}_{j_0}$ but you fit the full model, and $\mathbf{X}_{j_0}$ is nearly in $\mathcal{C}(\mathbf{X}_{-j_0})$. Then $(I - P_{-j_0})\mathbf{X}_{j_0} \approx \mathbf{0}$, so $d_{j_0} \approx 0$ and the *non-centrality* $\beta_{j_0} \sqrt{d_{j_0}}/\sigma$ is ≈ 0 : the statistic behaves like a central t_{n-p} and the test rejects at barely more than its level. In the true one-regressor model nothing is projected out, $d = \|\mathbf{X}_{j_0}\|^2$ is large, and the same β_{j_0} is highly significant.

$\hat{\beta}_{j_0}$ stays **unbiased** throughout. What collinearity inflates is $\text{Var}(\hat{\beta}_{j_0}) = \sigma^2/d_{j_0}$.

(c) **Analysis of covariance — the standard worked instance.** Model: r treatments plus two continuous covariates,

$$Y_{ij} = \mu_i + \beta_1 z_{ij} + \beta_2 w_{ij} + \varepsilon_{ij}, \quad j = 1, \dots, n_i, \quad i = 1, \dots, r, \quad n = \sum_i n_i.$$

Here $\mathbf{Z}$ is the $n \times r$ matrix of treatment indicators and $\mathbf{W} = [z : w]$.

The one observation that solves the whole problem

$P_{\mathbf{Z}}$ replaces each observation by its **treatment mean**, so $I - P_{\mathbf{Z}}$ **centres within treatment**: $[(I - P_{\mathbf{Z}})v]_{ij} = v_{ij} - \bar{v}_{i\cdot}$. Hence, by symmetry and idempotence,

$$u^T (I - P_{\mathbf{Z}}) v = \sum_i \sum_j (u_{ij} - \bar{u}_{i\cdot})(v_{ij} - \bar{v}_{i\cdot}).$$

Every “ S ” below is one of these inner products.

With $S_{zz} = \sum_i \sum_j (z_{ij} - \bar{z}_{i\cdot})^2$, and $S_{zw}, S_{ww}, S_{zy}, S_{wy}$ defined the same way, Frisch–Waugh reads

$$\mathbf{W}^T (I - P_{\mathbf{Z}}) \mathbf{W} = \begin{pmatrix} S_{zz} & S_{zw} \\ S_{zw} & S_{ww} \end{pmatrix} =: M, \quad \mathbf{W}^T (I - P_{\mathbf{Z}}) \mathbf{Y} = \begin{pmatrix} S_{zy} \\ S_{wy} \end{pmatrix},$$

ANCOVA estimates (2025 exam, 12 marks)

$$\hat{\beta} = M^{-1} \begin{pmatrix} S_{zy} \\ S_{wy} \end{pmatrix}, \quad \text{Var}(\hat{\beta}) = \sigma^2 M^{-1}, \quad \text{d.f.} = n - r - 2.$$

The two natural tests then follow from §6 (all of Wald, F and LRT coincide):

$$H_0 : \beta_2 = 0 : [M^{-1}]_{22} = \frac{S_{zz}}{S_{zz}S_{ww} - S_{zw}^2} = \left(S_{ww} - \frac{S_{zw}^2}{S_{zz}} \right)^{-1} \implies U = \left(S_{ww} - \frac{S_{zw}^2}{S_{zz}} \right) \frac{\hat{\beta}_2^2}{\hat{\sigma}^2} \sim F_{1, n-r-2},$$

$$H_0 : \beta_1 = \beta_2 = 0 : \quad V = \frac{1}{\hat{\sigma}^2} \hat{\beta}^T M \hat{\beta}, \quad \frac{V}{2} \sim F_{2, n-r-2},$$

both rejecting for *large* values.

Reading the coefficient in U

$S_{ww} - S_{zw}^2/S_{zz} = w^T(I - P_{[\mathbf{Z}; z]})w$ is what is left of w after removing *both* the treatment means and z . So U is $\hat{\beta}_2^2 \times (\text{information about } \beta_2)/\hat{\sigma}^2$ — the same collinearity story as (b): the more of w survives adjustment for z , the more powerful the test.

3 R^2 and partial correlation

3.1 R^2 as a ratio of projections

Total variability about the mean:

$$\text{SST} = \sum_i (Y_i - \bar{Y})^2 = \|\mathbf{Y}\|^2 - n\bar{Y}^2 = \mathbf{Y}^T \mathbf{Y} - \mathbf{Y}^T \left(\frac{1}{n} \mathbf{1} \mathbf{1}^T \right) \mathbf{Y} = \mathbf{Y}^T (I - P_1) \mathbf{Y}.$$

And $\text{SSE} = \mathbf{Y}^T (I - P_{\mathbf{X}}) \mathbf{Y}$, so $\text{SSR} = \mathbf{Y}^T (P_{\mathbf{X}} - P_1) \mathbf{Y}$. Hence

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{\mathbf{Y}^T (P_{\mathbf{X}} - P_1) \mathbf{Y}}{\mathbf{Y}^T (I - P_1) \mathbf{Y}}$$

— the proportion of variability in the response explained by the predictors. For $p = 2$ (simple linear regression) this reduces to $R^2 = (\text{corr}(x, Y))^2$:

$$R^2 = \frac{(\sum (x_i - \bar{x})(Y_i - \bar{Y}))^2}{\sum (x_i - \bar{x})^2 \sum (Y_i - \bar{Y})^2} = \frac{\hat{\beta}_1^2 \sum (x_i - \bar{x})^2}{\sum (Y_i - \bar{Y})^2} = \frac{\text{SSR}}{\text{SST}}.$$

3.2 Why R^2 never decreases when you add a predictor

$$R_{\text{full}}^2 \geq R_{\text{sub}}^2, \text{ always}$$

The denominators are identical, so compare numerators:

$$\mathbf{Y}^T (P_{\mathbf{X}} - P_1) \mathbf{Y} - \mathbf{Y}^T (P_{\mathbf{Z}} - P_1) \mathbf{Y} = \mathbf{Y}^T (P_{\mathbf{X}} - P_{\mathbf{Z}}) \mathbf{Y} = \mathbf{Y}^T P_{\mathbf{W}|\mathbf{Z}} \mathbf{Y} \geq 0$$

by the lemma, because $P_{\mathbf{W}|\mathbf{Z}}$ is a projection matrix and hence non-negative definite ($\mathbf{Y}^T P \mathbf{Y} = \|P \mathbf{Y}\|^2$).

Moral: R^2 is useless for choosing between nested models — it always favours the bigger one. That is precisely why you need the F -test, which penalises the extra parameters through the degrees of freedom.

3.3 Partial correlation

The partial correlation between Y and $X^{(j)}$ given the other predictors is the ordinary correlation between the two sets of leftovers:

$$e^{(-j)} = (I - H_{(-j)}) \mathbf{Y} \quad (\text{residuals of } Y \text{ on the others}), \quad r^{(-j).j} = (I - H_{(-j)}) X^{(j)} \quad (\text{residuals of } X^{(j)} \text{ on the others})$$

where $H_{(-j)}$ projects onto $\mathcal{C}([\mathbf{1} : X^{(1)} : \dots \widehat{X^{(j)}} \dots : X^{(p-1)}])$. Then

$$\rho_{YX^{(j)}|\{X^{(k)}\}_{k \neq j}} = \frac{(X^{(j)})^T(I - H_{(-j)})\mathbf{Y}}{\sqrt{\mathbf{Y}^T(I - H_{(-j)})\mathbf{Y}} \cdot \sqrt{(X^{(j)})^T(I - H_{(-j)})X^{(j)}}}.$$

Same idea as Frisch–Waugh: remove the common part first, then correlate.

4 Distribution theory of quadratic forms

This is the technical core. Everything reduces to one theorem.

4.1 MGFs identify distributions

$X \stackrel{d}{=} Y$ iff their characteristic functions agree. If the MGF $\psi_X(t) = \mathbb{E}e^{tX}$ is finite in an open neighbourhood of 0, then $\phi_X(t) = \psi_X(it)$ there and the MGF determines the law. So computing an MGF and recognising it is a legitimate proof technique.

4.2 The chi-square MGF

$\chi_n^2 = Z_1^2 + \dots + Z_n^2$ with $Z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. Since $\chi_2^2 \sim \text{Exp}(1/2)$ with density $\frac{1}{2}e^{-x/2}$,

$$\psi_2(t) = \int_0^\infty e^{tx} \frac{1}{2}e^{-x/2} dx = \frac{1}{1-2t} \quad (t < \tfrac{1}{2}).$$

By independence $\psi_2 = \psi_1^2$, so $\psi_1(t) = (1-2t)^{-1/2}$ and

$$\boxed{\psi_{\chi_n^2}(t) = (1-2t)^{-n/2}, \quad t < \tfrac{1}{2}.$$

4.3 The MGF of a general quadratic form

Master formula

If $\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, I_n)$ and A is symmetric non-negative definite, then for t such that $(I - 2tA)^{-1}$ exists,

$$\mathbb{E}[e^{t\mathbf{Y}^T A \mathbf{Y}}] = |\det(I - 2tA)|^{-1/2} \exp\left(-\tfrac{1}{2}\boldsymbol{\mu}^T \boldsymbol{\mu} + \tfrac{1}{2}\boldsymbol{\mu}^T (I - 2tA)^{-1} \boldsymbol{\mu}\right).$$

(Obtained by writing out the $\mathcal{N}(\boldsymbol{\mu}, I)$ density and completing the square in the exponent; the det is the Jacobian of the resulting Gaussian integral.)

4.4 The key theorem

Theorem (quadratic forms and idempotence)

Let $\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, I)$ and A symmetric. Then

$$\mathbf{Y}^T A \mathbf{Y} \sim \chi_{\rho(A)}^2\left(\tfrac{1}{2}\boldsymbol{\mu}^T A \boldsymbol{\mu}\right) \iff A \text{ is idempotent.}$$

(The argument in parentheses is the *non-centrality parameter*; it is 0 — giving a central χ^2 — exactly when $A\boldsymbol{\mu} = 0$.)

Proof. (central case $\boldsymbol{\mu} = \mathbf{0}$, both directions). Spectrally decompose A : eigenvalues $\lambda_1, \dots, \lambda_r$ non-zero and the rest 0, where $r = \rho(A)$. Then

$$\psi(t) = |\det(I - 2tA)|^{-1/2} = \left(\prod_{j=1}^r (1 - 2t\lambda_j) \right)^{-1/2}.$$

This equals the χ_r^2 MGF $(1 - 2t)^{-r/2}$ for all t near 0 iff

$$(1-2t)^r = \prod_{j=1}^r (1-2t\lambda_j) \quad \forall t \iff \lambda_j = 1 \quad \forall j \iff A \text{ has only eigenvalues } 0, 1 \iff A^2 = A. \quad \square$$

The corollary you actually use

$\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, \sigma^2 I)$ and A symmetric idempotent with $A\boldsymbol{\mu} = 0 \implies \frac{\mathbf{Y}^T A \mathbf{Y}}{\sigma^2} \sim \chi_{\rho(A)}^2$.
 Applied with $A = I - P_{\mathbf{X}}$ and $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ (so $A\boldsymbol{\mu} = 0$): $\text{SSE} \sim \sigma^2 \chi_{n-\rho(\mathbf{X})}^2$.

4.5 Fisher–Cochran theorem

Fisher–Cochran

Let $\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, I)$ and $A_1, \dots, A_r$ be symmetric non-negative definite with

$$\mathbf{Y}^T \mathbf{Y} = \sum_{j=1}^r \mathbf{Y}^T A_j \mathbf{Y}.$$

Then $\mathbf{Y}^T A_j \mathbf{Y} \sim \chi_{\rho(A_j)}^2 (\lambda_j/2)$ for each j and these forms are mutually independent $\iff$

$$\sum_{j=1}^r \rho(A_j) = n.$$

In that case $\lambda_j = \boldsymbol{\mu}^T A_j \boldsymbol{\mu}$ and $\sum_j \lambda_j = \boldsymbol{\mu}^T \boldsymbol{\mu}$. Moreover each A_j is idempotent and $A_j A_k = 0$ for $j \neq k$.

Proof. ($\Leftarrow$, the direction you use) Rank-factorise $A_j = C_j C_j^T$ with C_j of full column rank $\rho(A_j)$. Then

$$\mathbf{Y}^T \mathbf{Y} = \sum_j \mathbf{Y}^T C_j C_j^T \mathbf{Y} = \mathbf{Y}^T B B^T \mathbf{Y}, \quad B = [C_1 : \dots : C_r].$$

By assumption B is $n \times n$. Since $\mathbf{Y}^T B B^T \mathbf{Y} = \mathbf{Y}^T \mathbf{Y}$ for all $\mathbf{Y}$, spectral decomposition forces every eigenvalue of $B B^T$ to be 1, so $B B^T = I_n$ and hence $B^T B = I_n$ — B is orthogonal. Reading off the blocks of $B^T B = I_n$:

$$C_j^T C_j = I_{\rho(A_j)}, \quad C_j^T C_k = 0 \quad (j \neq k).$$

So $A_j^2 = C_j C_j^T C_j C_j^T = C_j I C_j^T = A_j$ (idempotent, hence each form is χ^2 by the previous theorem), and $A_j A_k = C_j (C_j^T C_k) C_k^T = 0$. Independence follows because $U := B^T \mathbf{Y} \sim \mathcal{N}_n(B^T \boldsymbol{\mu}, I)$ has independent coordinates and $\mathbf{Y}^T A_j \mathbf{Y}$ is the sum of squares of the block of U belonging to C_j — disjoint blocks, hence independent. ■

What Fisher–Cochran is for

It is the theorem that legitimises the **ANOVA table**. It says: if you split $\|\mathbf{Y}\|^2$ (or SST) into pieces whose degrees of freedom add up correctly, then those pieces are automatically

independent chi-squares — so their ratio is an F . The d.f. column of the ANOVA table adding up is not bookkeeping, it is the **hypothesis** of the theorem.

5 Constrained least squares

5.1 The general null hypothesis

Any hypothesis specifying a sub-model can be written

$$H_0 : \underset{r \times p}{L^T} \underset{r \times 1}{\beta} = \underset{r \times 1}{\theta_0}, \quad \rho(\mathbf{X}) = p, \quad \rho(L) = r \leq p.$$

Examples. With $p = 4$ and $\beta = (\beta_0, \dots, \beta_3)$:

$$H_0 : \begin{cases} \beta_1 = \beta_2 \\ \beta_2 + \beta_3 = 0 \end{cases} \iff L^T = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad \theta_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and under it the mean becomes $\beta_0 + \beta_1(X^{(1)} + X^{(2)} - X^{(3)})$ — a model with one predictor instead of three. If instead $\beta_2 + \beta_3 = 1$, the mean is $\beta_0 + \beta_1(X^{(1)} + X^{(2)} - X^{(3)}) + X^{(3)}$, with a known *offset*.

5.2 Solving the constrained problem

$$\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|^2 \quad \text{subject to} \quad L^T \beta = \theta_0.$$

Lagrangian $\mathcal{L}(\beta, \lambda) = \frac{1}{2} \|\mathbf{Y} - \mathbf{X}\beta\|^2 + \lambda^T (L^T \beta - \theta_0)$.

Trick: whiten first. Put $\mathbf{X}_0 = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1/2}$ so that $\mathbf{X}_0^T \mathbf{X}_0 = I_p$, and $\gamma = (\mathbf{X}^T \mathbf{X})^{1/2} \beta$, $L_0 = (\mathbf{X}^T \mathbf{X})^{-1/2} L$. Then $\mathbf{X}\beta = \mathbf{X}_0 \gamma$ and $L^T \beta = L_0^T \gamma$, so the problem becomes $\min \frac{1}{2} \|\mathbf{Y} - \mathbf{X}_0 \gamma\|^2$ s.t. $L_0^T \gamma = \theta_0$. Stationarity:

$$-\mathbf{X}_0^T (\mathbf{Y} - \mathbf{X}_0 \gamma) + L_0 \lambda = 0 \implies \gamma + L_0 \lambda = \mathbf{X}_0^T \mathbf{Y} \quad (\text{i})$$

$$L_0^T \gamma = \theta_0 \quad (\text{ii})$$

Multiply (i) by L_0^T and use (ii): $\theta_0 + L_0^T L_0 \lambda = L_0^T \mathbf{X}_0^T \mathbf{Y}$, so $\lambda = (L_0^T L_0)^{-1} (L_0^T \mathbf{X}_0^T \mathbf{Y} - \theta_0)$. Substituting back and noting that the unconstrained estimate is $\hat{\gamma} = \mathbf{X}_0^T \mathbf{Y}$:

$$\tilde{\gamma} = \hat{\gamma} - L_0 (L_0^T L_0)^{-1} (\hat{\theta} - \theta_0), \quad \hat{\theta} := L^T \hat{\beta}.$$

Translating back:

Constrained LSE

$$\tilde{\beta} = \hat{\beta} - (\mathbf{X}^T \mathbf{X})^{-1} L (L^T (\mathbf{X}^T \mathbf{X})^{-1} L)^{-1} (\hat{\theta} - \theta_0), \quad \hat{\theta} = L^T \hat{\beta}.$$

Read it as: *start at the unconstrained estimate, then step back by exactly the amount needed to satisfy the constraint*, in the metric $(\mathbf{X}^T \mathbf{X})^{-1}$.

5.3 The reduction in SSE

The identity that makes the F -test work

$$\text{SSE}_{\text{red}} - \text{SSE}_{\text{full}} = \|\mathbf{X}\hat{\beta} - \mathbf{X}\tilde{\beta}\|^2 = (\hat{\theta} - \theta_0)^T (L^T (\mathbf{X}^T \mathbf{X})^{-1} L)^{-1} (\hat{\theta} - \theta_0).$$

Proof. $\|\mathbf{Y} - \mathbf{X}\tilde{\beta}\|^2 = \|(\mathbf{Y} - P_{\mathbf{X}} \mathbf{Y}) + (P_{\mathbf{X}} \mathbf{Y} - \mathbf{X}\tilde{\beta})\|^2$; the two terms are orthogonal ($P_{\mathbf{X}} \mathbf{Y} - \mathbf{X}\tilde{\beta} \in \mathcal{C}(\mathbf{X})$, the other in $\mathcal{C}(\mathbf{X})^\perp$), so $\text{SSE}_{\text{red}} = \text{SSE}_{\text{full}} + \|\mathbf{X}\hat{\beta} - \mathbf{X}\tilde{\beta}\|^2$. Then substitute the boxed formula for $\tilde{\beta}$ and simplify. ■

6 The three equivalent tests

Model: $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 I)$, $\rho(\mathbf{X}) = p$; $H_0 : L^T \boldsymbol{\beta} = \theta_0$ vs $H_1 : L^T \boldsymbol{\beta} \neq \theta_0$, with $\rho(L) = r$.

6.1 Wald's test (based on the BLUE)

$\hat{\boldsymbol{\theta}} = L^T \hat{\boldsymbol{\beta}} \sim \mathcal{N}_r(L^T \boldsymbol{\beta}, \sigma^2 L^T (\mathbf{X}^T \mathbf{X})^{-1} L)$, and $\hat{\sigma}^2 = \text{MSE} = \mathbf{Y}^T (I - P_{\mathbf{X}}) \mathbf{Y} / (n - p)$ independently. Define

$$W = (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)^T (\widehat{\text{Var}}(\hat{\boldsymbol{\theta}}))^{-1} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \frac{1}{\hat{\sigma}^2} (L^T \hat{\boldsymbol{\beta}} - \boldsymbol{\theta}_0)^T (L^T (\mathbf{X}^T \mathbf{X})^{-1} L)^{-1} (L^T \hat{\boldsymbol{\beta}} - \boldsymbol{\theta}_0).$$

Distribution

$$\frac{W}{r} \sim F_{r, n-p} \left(\frac{\lambda}{2} \right), \quad \lambda = \frac{1}{\sigma^2} (L^T \boldsymbol{\beta} - \boldsymbol{\theta}_0)^T (L^T (\mathbf{X}^T \mathbf{X})^{-1} L)^{-1} (L^T \boldsymbol{\beta} - \boldsymbol{\theta}_0).$$

Under H_0 , $\lambda = 0$ and $W/r \sim F_{r, n-p}$ exactly. The power increases to 1 as $\lambda \rightarrow \infty$ — i.e. the further the truth is from H_0 (measured in the right metric), the more surely you reject.

6.2 The F -test (based on the drop in SSE)

By §5.3 the numerator of W is exactly $\text{SSE}_{\text{red}} - \text{SSE}_{\text{full}}$, and the reduced model has $p - r$ free parameters, so it has $n - (p - r)$ residual d.f. Hence

The F -statistic — the form you will actually compute

$$F = \frac{(\text{SSE}_{\text{red}} - \text{SSE}_{\text{full}}) / [\text{df}_{\text{red}} - \text{df}_{\text{full}}]}{\text{MSE}_{\text{full}}} = \frac{W}{(n - (p - r)) - (n - p)} = \frac{W}{r} \sim F_{r, n-p} \text{ under } H_0.$$

Wald's test and the F -test are literally the same number. Reject if $F > F_{r, n-p}^{1-\alpha}$.

In the sub-model language of §2, testing $H_0 : \boldsymbol{\delta} = \mathbf{0}$ uses

$$F = \frac{\mathbf{Y}^T P_{\mathbf{W}|\mathbf{Z}} \mathbf{Y} / s}{\mathbf{Y}^T (I - P_{\mathbf{X}}) \mathbf{Y} / (n - p)} = \frac{\mathbf{Y}^T P_{\mathbf{W}|\mathbf{Z}} \mathbf{Y} / s}{\text{MSE}},$$

with the ANOVA decomposition

$$\mathbf{Y}^T \mathbf{Y} = \underbrace{\mathbf{Y}^T P_{\mathbf{Z}} \mathbf{Y}}_{\text{explained by } \mathbf{Z}} + \underbrace{\mathbf{Y}^T P_{\mathbf{W}|\mathbf{Z}} \mathbf{Y}}_{\text{extra from } \mathbf{W}} + \underbrace{\mathbf{Y}^T (I - P_{\mathbf{X}}) \mathbf{Y}}_{\text{SSE}},$$

whose three ranks are q , s , $n - p$ and sum to n — Fisher–Cochran applies, giving independence and the F .

6.3 The likelihood ratio test

$\Theta = \mathbb{R}^p \times \mathbb{R}_+$, $\Theta_0 = \{\boldsymbol{\beta} : L^T \boldsymbol{\beta} = \boldsymbol{\theta}_0\} \times \mathbb{R}_+$, and

$$\Lambda = \frac{\sup_{\Theta_0} \ell(\boldsymbol{\theta} \mid \text{data})}{\sup_{\Theta} \ell(\boldsymbol{\theta} \mid \text{data})}.$$

Denominator: for fixed σ^2 the maximiser over $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}}$; then maximising over σ^2 gives $\hat{\sigma}^2 = \frac{1}{n} \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2$, and the maximum equals $(\hat{\sigma}^2)^{-n/2} e^{-n/2}$ up to the constant. Numerator: the same with $\hat{\boldsymbol{\beta}}$

replaced by $\tilde{\beta}$ and $\tilde{\sigma}^2 = \frac{1}{n} \|\mathbf{Y} - \mathbf{X}\tilde{\beta}\|^2$. Therefore

$$\Lambda = \left(\frac{\|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2}{\|\mathbf{Y} - \mathbf{X}\tilde{\beta}\|^2} \right)^{n/2} = \left(\frac{\text{SSE}_{\text{full}}}{\text{SSE}_{\text{red}}} \right)^{n/2}.$$

The LRT is the F -test in disguise

Reject for small Λ , i.e. for large

$$\frac{\text{SSE}_{\text{red}}}{\text{SSE}_{\text{full}}} - 1 = \frac{\text{SSE}_{\text{red}} - \text{SSE}_{\text{full}}}{\text{SSE}_{\text{full}}},$$

which is a strictly increasing function of F . So the LRT, Wald's test and the F -test give *identical* rejection regions. (Asymptotically, $-2 \log \Lambda \xrightarrow{d} \chi_r^2$; here we have the exact finite-sample F instead, which is better.)

7 The fundamental theorems of least squares

Model $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\beta, \sigma^2 I)$, $\rho(\mathbf{X}) = r \leq p$, $\hat{\beta}$ any solution of $\mathbf{X}^T \mathbf{X} \beta = \mathbf{X}^T \mathbf{Y}$, and $\text{SSE} = \mathbf{Y}^T (I - P_{\mathbf{X}}) \mathbf{Y}$.

First fundamental theorem

$$\text{SSE} = \min_{\beta \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\beta\|^2, \quad \text{SSE} \sim \sigma^2 \chi_{n-r}^2.$$

Second fundamental theorem

Let L be $p \times m$ with $\mathcal{C}(L) \subseteq \mathcal{C}(\mathbf{X}^T)$ (i.e. every component of $L^T \beta$ is estimable), and $\text{SSE}_{\text{red}} = \min_{\beta: L^T \beta = \theta_0} \|\mathbf{Y} - \mathbf{X}\beta\|^2$. Then

- (i) SSE and $\text{SSE}_{\text{red}} - \text{SSE}$ are **independent**;
- (ii) $\text{SSE}_{\text{red}} - \text{SSE} \sim \sigma^2 \chi_m^2(\delta)$ for some non-centrality δ , with $\delta = 0$ precisely when $L^T \beta = \theta_0$.

This theorem *is* the justification of the F -test: (i) gives independence of numerator and denominator, (ii) gives the null distribution and shows the test has power against every alternative.

Third fundamental theorem

Let $\mathbf{X} = [\mathbf{X}_1 : \mathbf{X}_2]$ with $\mathbf{X}_1$ of q columns and $r_1 = \rho(\mathbf{X}_1)$. Define

$$U = \frac{\min_{\beta \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\beta\|^2}{\min_{\gamma \in \mathbb{R}^q} \|\mathbf{Y} - \mathbf{X}_1 \gamma\|^2} = \frac{\text{SSE}_{\text{full}}}{\text{SSE}_{\text{red}}}.$$

Then $U \sim \text{Beta}\left(\frac{n-r}{2}, \frac{r-r_1}{2}\right)$.

Proof. $S_1 := \text{SSE}_{\text{full}} \sim \sigma^2 \chi_{n-r}^2$ and $S_0 - S_1 = \mathbf{Y}^T (P_{\mathbf{X}} - P_{\mathbf{X}_1}) \mathbf{Y} \sim \sigma^2 \chi_{r-r_1}^2$, and they are independent because $(I - P_{\mathbf{X}})(P_{\mathbf{X}} - P_{\mathbf{X}_1}) = O$. Both are *central* precisely when $\mathbb{E}\mathbf{Y} \in \mathcal{C}(\mathbf{X}_1)$, i.e. when the sub-model holds. Then

$$U = \frac{S_1}{S_0} = \frac{S_1}{S_1 + (S_0 - S_1)} = \frac{A}{A+B}, \quad A \sim \chi_{n-r}^2 \text{ independent of } B \sim \chi_{r-r_1}^2,$$

and $A/(A+B) \sim \text{Beta}(a/2, b/2)$ for independent χ_a^2, χ_b^2 (change variables to $(A/(A+B), A+B)$; the Jacobian factorises the joint Gamma density). ■

Common confusion: which way up is the ratio?

$S_1 \leq S_0$ always — the bigger model cannot fit worse — so the Beta variate is $\text{SSE}_{\text{full}}/\text{SSE}_{\text{red}} \in [0, 1]$, *full over reduced*. Writing it the other way gives a quantity ≥ 1 , which cannot be Beta. That inequality is the fastest check, and it also tells you the first parameter is half the *numerator's* degrees of freedom. In the 2025 paper this appeared verbatim as $S_1/S_0 \sim \text{Beta}(\frac{n-p}{2}, \frac{p-q}{2})$ with $p = \rho(\mathbf{X})$, $q = \rho(\mathbf{Z})$.

8 Practice, with answers

- P1.** Show that if A is symmetric idempotent of rank k then $\text{tr}(A) = k$, and use it to write down $\mathbb{E}[\mathbf{Y}^T A \mathbf{Y}]$ when $\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, \sigma^2 I)$.
- P2.** In the sub-model setup, verify directly that $\rho(P_{\mathbf{Z}}) + \rho(P_{\mathbf{W}|\mathbf{Z}}) + \rho(I - P_{\mathbf{X}}) = n$, and say why this matters.
- P3.** Testing $H_0 : \beta_1 = 0$ in simple linear regression, show that the F -statistic equals T_1^2 where T_1 is the t -statistic of Crash Course 1.
- P4.** $n = 20$, $p = 5$. $\text{SSE}_{\text{full}} = 45$, and imposing 2 linear constraints gives $\text{SSE}_{\text{red}} = 63$. Carry out the F -test at 5% given $F_{2,15}^{0.95} = 3.68$.
- P5.** Explain why R^2 increasing when you add a predictor is *not* evidence that the predictor is useful, and state what the F -test adds.

Answers

- P1.** Idempotence forces eigenvalues in $\{0, 1\}$; the number of 1s is the rank k ; trace = sum of eigenvalues = k . Then $\mathbb{E}[\mathbf{Y}^T A \mathbf{Y}] = \text{tr}(A \text{Var } \mathbf{Y}) + \boldsymbol{\mu}^T A \boldsymbol{\mu} = \sigma^2 k + \boldsymbol{\mu}^T A \boldsymbol{\mu}$. (This is HW1 Q1(a).)
- P2.** $\rho(P_{\mathbf{Z}}) = q$, $\rho(I - P_{\mathbf{X}}) = n - p$, and by the lemma $P_{\mathbf{W}|\mathbf{Z}} = P_{\mathbf{X}} - P_{\mathbf{Z}}$ so $\rho(P_{\mathbf{W}|\mathbf{Z}}) = p - q = s$. Sum = $q + s + (n - p) = n$. It matters because that sum being n is precisely the hypothesis of Fisher–Cochran, which is what delivers the independence of the three sums of squares — and without independence the ratio is not an F .
- P3.** Here $r = 1$, $L^T = (0, 1)$, $\theta_0 = 0$, $p = 2$. Then $L^T(\mathbf{X}^T \mathbf{X})^{-1} L$ is the $(2, 2)$ entry of $(\mathbf{X}^T \mathbf{X})^{-1}$, namely $1/S_x^2$. So

$$F = W = \frac{\hat{\beta}_1^2}{\hat{\sigma}^2/S_x^2} = \left(\frac{\hat{\beta}_1}{s(\hat{\beta}_1)} \right)^2 = T_1^2.$$

Consistently, $t_{n-2}^2 = F_{1,n-2}$ as distributions.

P4. $\text{df}_{\text{full}} = n - p = 15$, $\text{MSE}_{\text{full}} = 45/15 = 3$. Numerator = $(63 - 45)/2 = 9$. So $F = 9/3 = 3.0$. Since $3.0 < 3.68$, **do not reject** H_0 : the two constraints are consistent with the data at the 5% level.

P5. $R_{\text{full}}^2 \geq R_{\text{sub}}^2$ is an algebraic identity ($\mathbf{Y}^T P_{\mathbf{W}|\mathbf{Z}} \mathbf{Y} \geq 0$) that holds for *every* dataset, including pure noise — adding a column of random numbers always raises R^2 . It therefore carries no evidential content. The F -test measures the same increase but divides it by the number of extra parameters and calibrates against MSE, so it asks the right question: is the increase larger than chance alone would produce?

Companions: Simple Linear Regression; Matrix Algebra and the Gauss–Markov Framework; One-Way ANOVA.